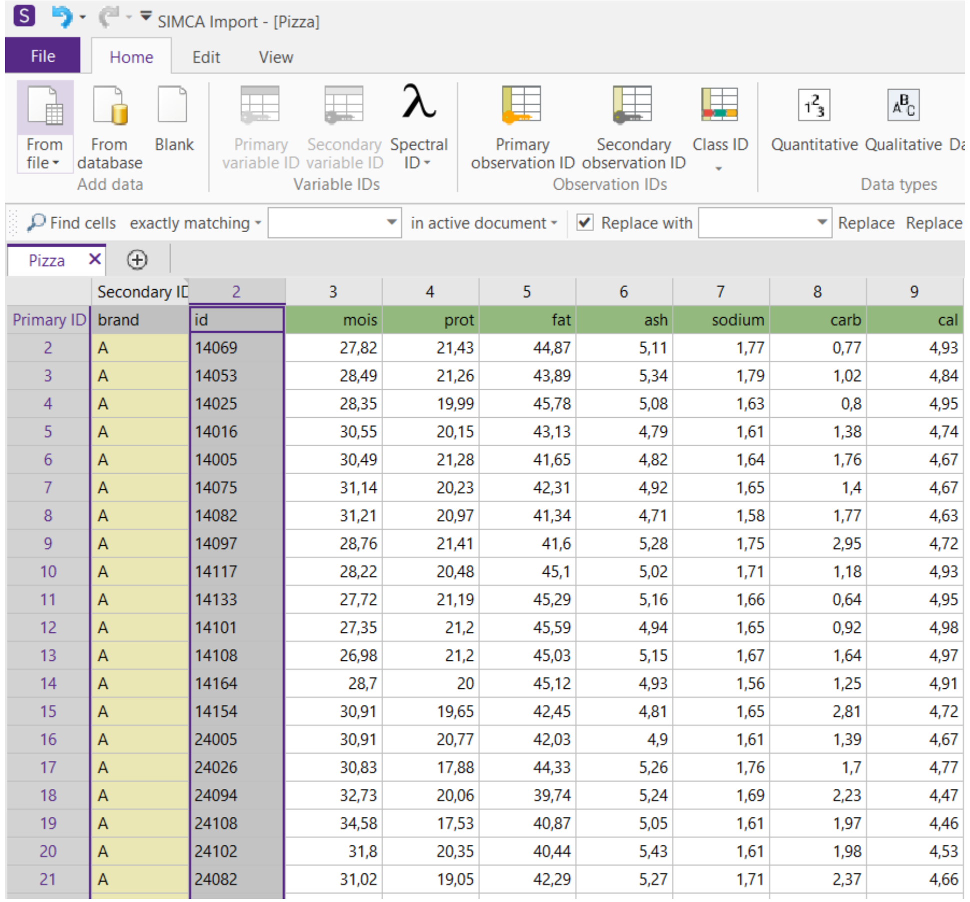
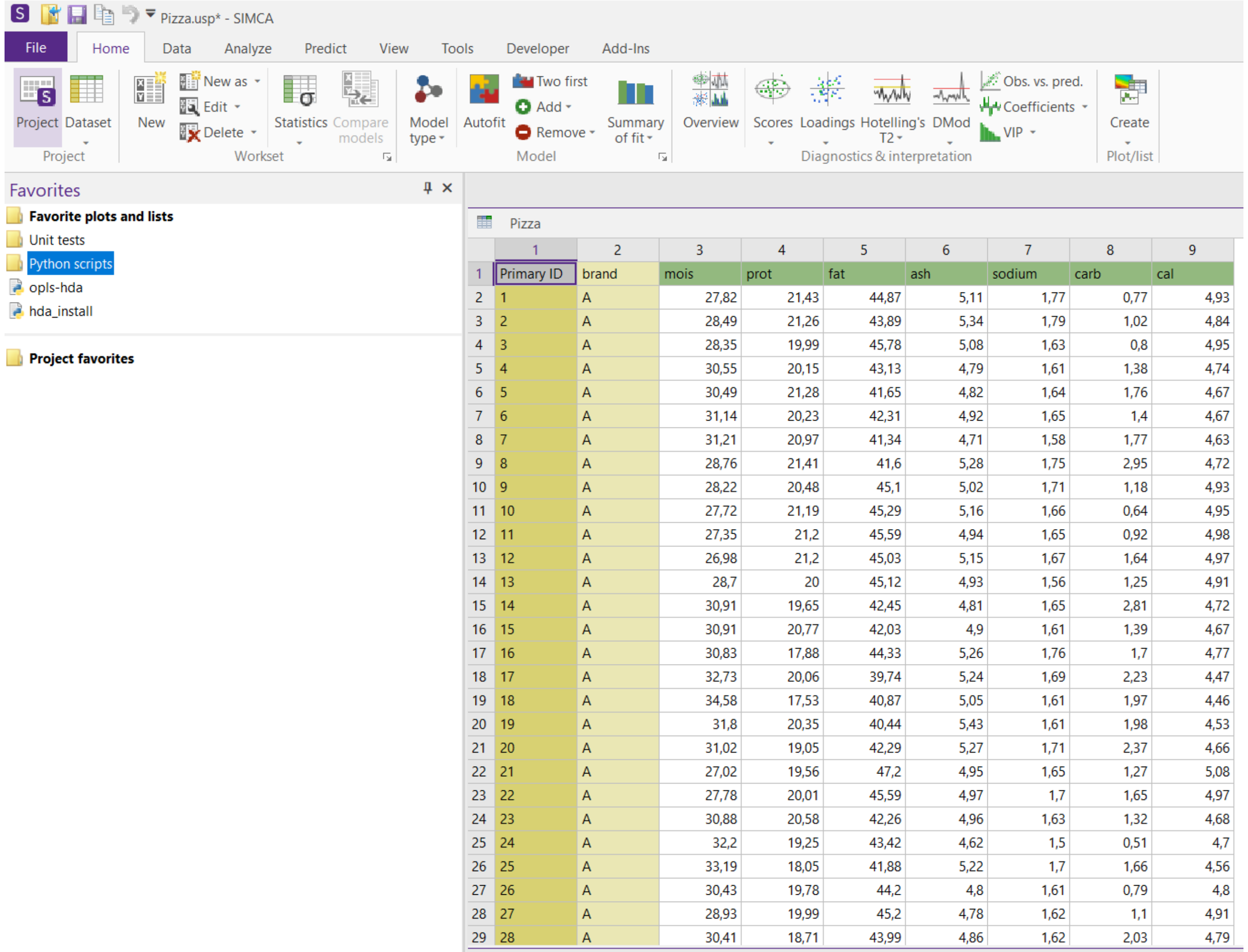
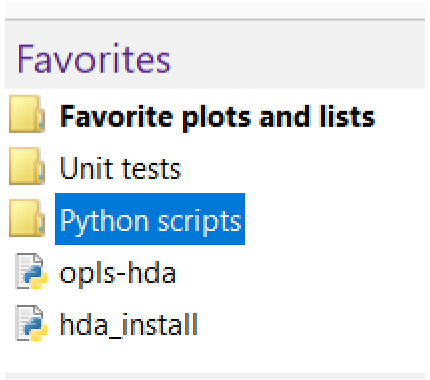
During import, include only the variables you intend to use. There is no include/exclude option available in the OPLS-HDA script. Set all observation IDs to Secondary. Simca will automatically generate a Primary ID (1 to N).



	Secondary ID	2	3	4	5	6	7	8	9
Primary ID	brand	id	mois	prot	fat	ash	sodium	carb	cal
2	A	14069	27,82	21,43	44,87	5,11	1,77	0,77	4,93
3	A	14053	28,49	21,26	43,89	5,34	1,79	1,02	4,84
4	A	14025	28,35	19,99	45,78	5,08	1,63	0,8	4,95
5	A	14016	30,55	20,15	43,13	4,79	1,61	1,38	4,74
6	A	14005	30,49	21,28	41,65	4,82	1,64	1,76	4,67
7	A	14075	31,14	20,23	42,31	4,92	1,65	1,4	4,67
8	A	14082	31,21	20,97	41,34	4,71	1,58	1,77	4,63
9	A	14097	28,76	21,41	41,6	5,28	1,75	2,95	4,72
10	A	14117	28,22	20,48	45,1	5,02	1,71	1,18	4,93
11	A	14133	27,72	21,19	45,29	5,16	1,66	0,64	4,95
12	A	14101	27,35	21,2	45,59	4,94	1,65	0,92	4,98
13	A	14108	26,98	21,2	45,03	5,15	1,67	1,64	4,97
14	A	14164	28,7	20	45,12	4,93	1,56	1,25	4,91
15	A	14154	30,91	19,65	42,45	4,81	1,65	2,81	4,72
16	A	24005	30,91	20,77	42,03	4,9	1,61	1,39	4,67
17	A	24026	30,83	17,88	44,33	5,26	1,76	1,7	4,77
18	A	24094	32,73	20,06	39,74	5,24	1,69	2,23	4,47
19	A	24108	34,58	17,53	40,87	5,05	1,61	1,97	4,46
20	A	24102	31,8	20,35	40,44	5,43	1,61	1,98	4,53
21	A	24082	31,02	19,05	42,29	5,27	1,71	2,37	4,66



	1	2	3	4	5	6	7	8	9
1	Primary ID	brand	mois	prot	fat	ash	sodium	carb	cal
2	1	A	27,82	21,43	44,87	5,11	1,77	0,77	4,93
3	2	A	28,49	21,26	43,89	5,34	1,79	1,02	4,84
4	3	A	28,35	19,99	45,78	5,08	1,63	0,8	4,95
5	4	A	30,55	20,15	43,13	4,79	1,61	1,38	4,74
6	5	A	30,49	21,28	41,65	4,82	1,64	1,76	4,67
7	6	A	31,14	20,23	42,31	4,92	1,65	1,4	4,67
8	7	A	31,21	20,97	41,34	4,71	1,58	1,77	4,63
9	8	A	28,76	21,41	41,6	5,28	1,75	2,95	4,72
10	9	A	28,22	20,48	45,1	5,02	1,71	1,18	4,93
11	10	A	27,72	21,19	45,29	5,16	1,66	0,64	4,95
12	11	A	27,35	21,2	45,59	4,94	1,65	0,92	4,98
13	12	A	26,98	21,2	45,03	5,15	1,67	1,64	4,97
14	13	A	28,7	20	45,12	4,93	1,56	1,25	4,91
15	14	A	30,91	19,65	42,45	4,81	1,65	2,81	4,72
16	15	A	30,91	20,77	42,03	4,9	1,61	1,39	4,67
17	16	A	30,83	17,88	44,33	5,26	1,76	1,7	4,77
18	17	A	32,73	20,06	39,74	5,24	1,69	2,23	4,47
19	18	A	34,58	17,53	40,87	5,05	1,61	1,97	4,46
20	19	A	31,8	20,35	40,44	5,43	1,61	1,98	4,53
21	20	A	31,02	19,05	42,29	5,27	1,71	2,37	4,66
22	21	A	27,02	19,56	47,2	4,95	1,65	1,27	5,08
23	22	A	27,78	20,01	45,59	4,97	1,7	1,65	4,97
24	23	A	30,88	20,58	42,26	4,96	1,63	1,32	4,68
25	24	A	32,2	19,25	43,42	4,62	1,5	0,51	4,7
26	25	A	33,19	18,05	41,88	5,22	1,7	1,66	4,56
27	26	A	30,43	19,78	44,2	4,8	1,61	0,79	4,8
28	27	A	28,93	19,99	45,2	4,78	1,62	1,1	4,91
29	28	A	30,41	18,71	43,99	4,86	1,62	2,03	4,79

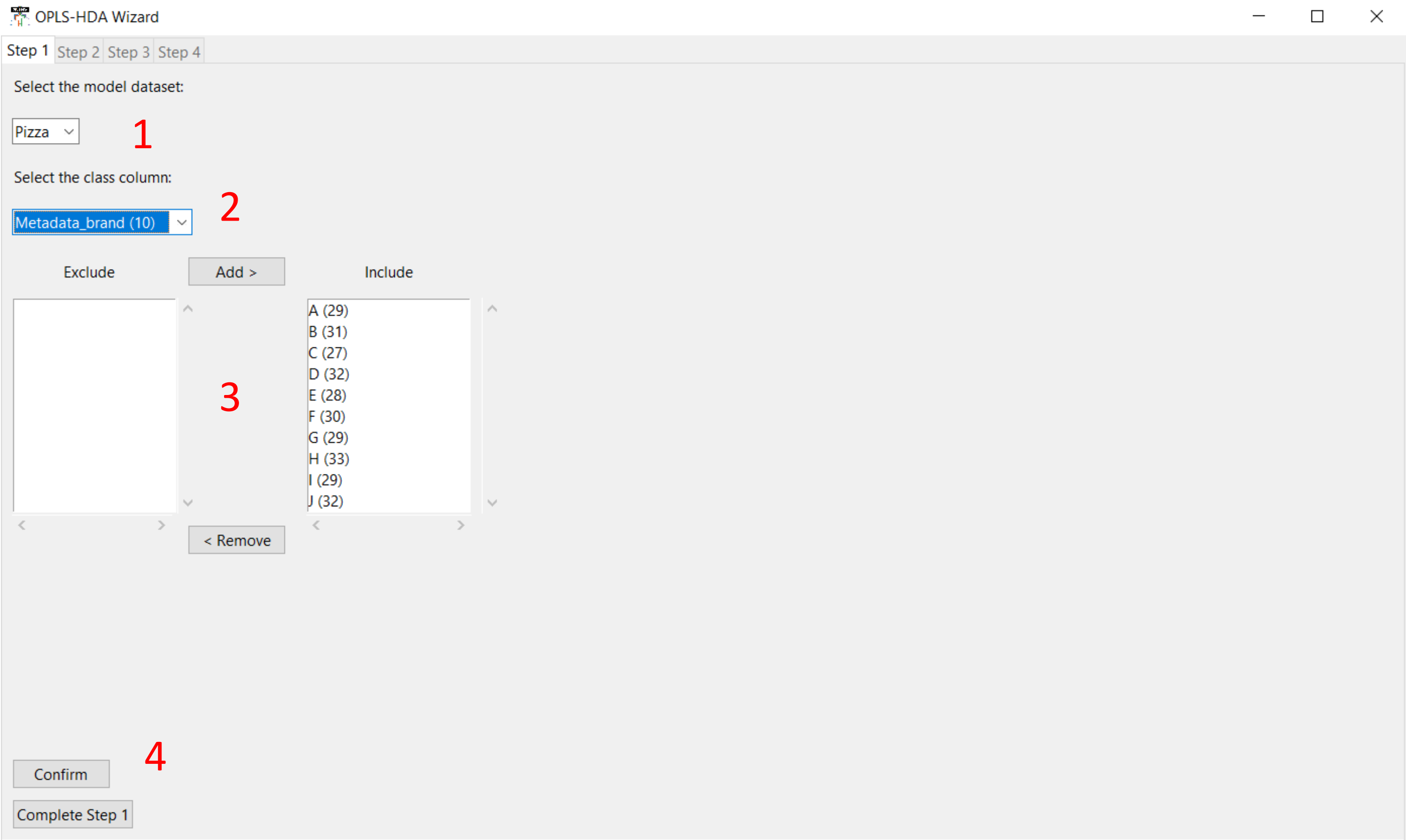


Install the script and python packages

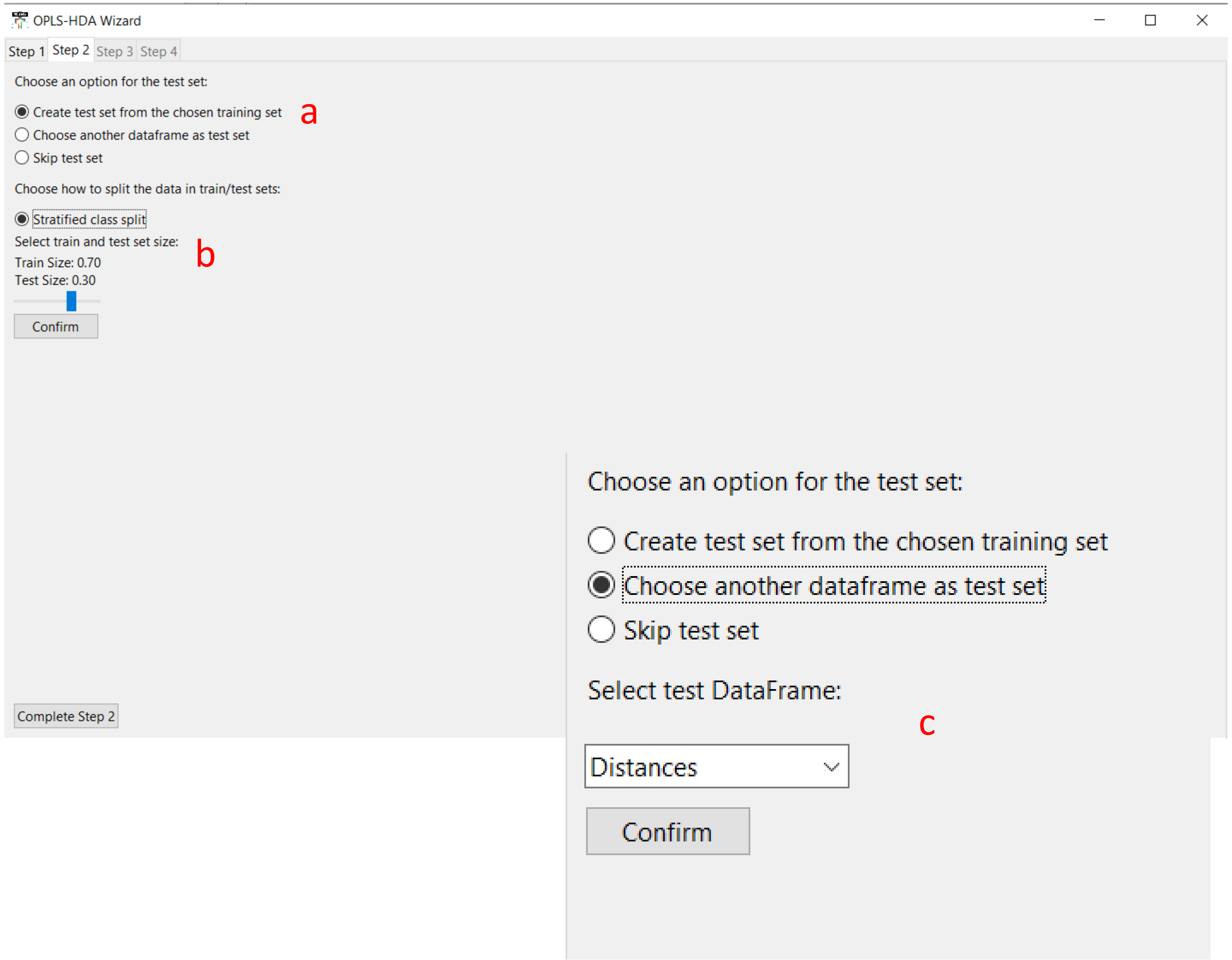
Drag the files opls-hda.py and hda_install.py to Favorites in Simca.
Click on "hda_install" to install the required Python packages.
Then, close and restart Simca.

Run the OPLS-HDA script

Click “opls-hda”



1) Select the dataset you want to use for training the model.
2) Choose the column in the dataset that contains the class information. You can select from a list of columns that you set as Secondary ID during import.
3)
Use "Add >" and "< Remove" to select the classes to be included in the OPLS-HDA model.
4) When finished, click "Confirm" and then "Complete Step 1."



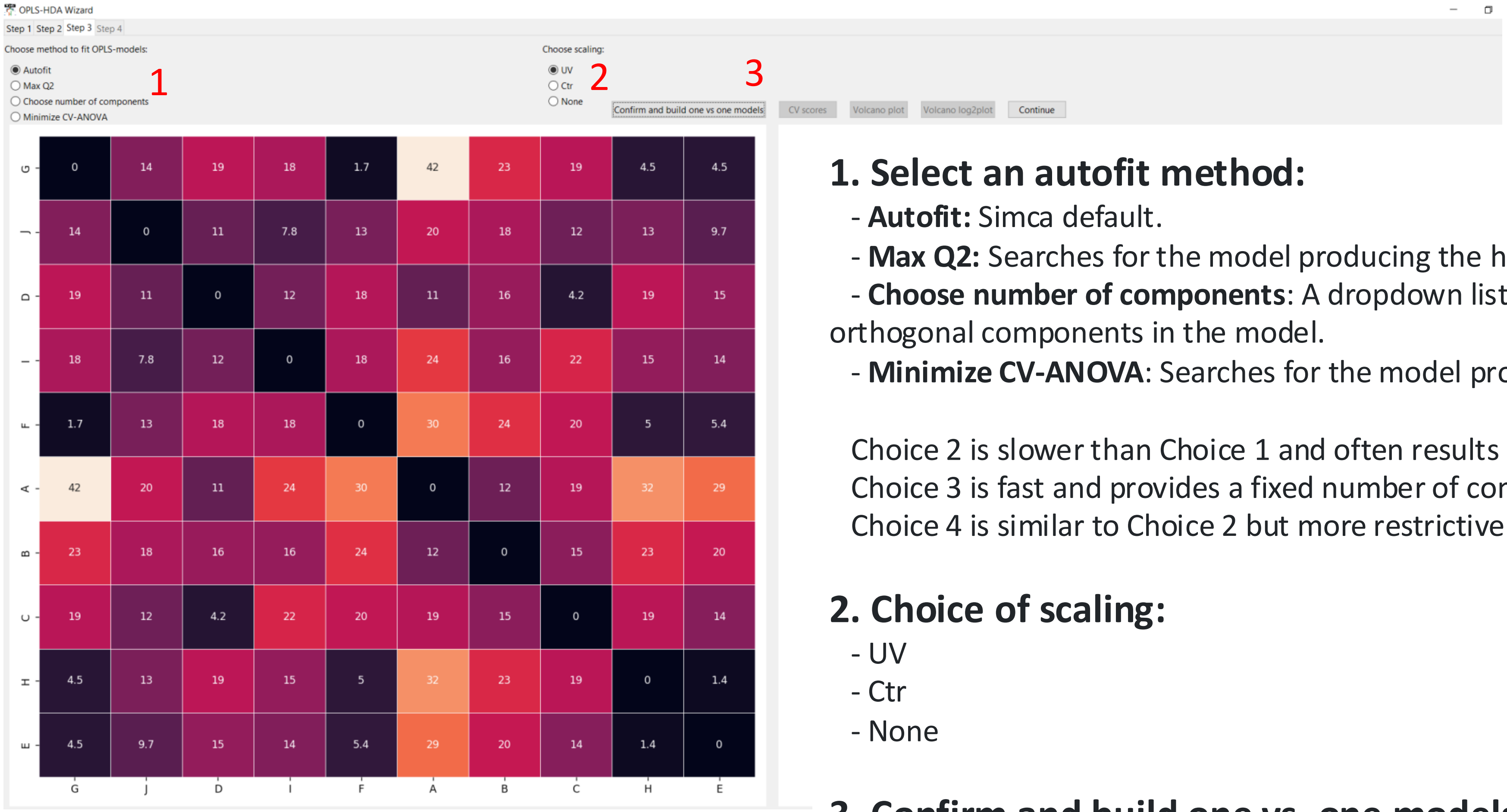
Select a test set. There are three options: (“a”)

1.Create test set from the chosen training set: The training set will be split into two parts— a training set and a test set. Select the proportion of the split using "stratified class split“ (“b”). A random split will divide each class in the given proportions.

2.Choose another dataframe as test set: Choose a dataset from the dropdown list (“c”). It needs to have the same variables and observation IDs as your training set.

3.Skip test set: No test set will be used.

When finished, click "Confirm" and then "Complete Step 2."



1. Select an autofit method:

- **Autofit:** Simca default.
- **Max Q2:** Searches for the model producing the highest Q2 value.
- **Choose number of components**: A dropdown list appears, allowing you to set the number of orthogonal components in the model.
- **Minimize CV-ANOVA**: Searches for the model producing the lowest CV-ANOVA p-value.

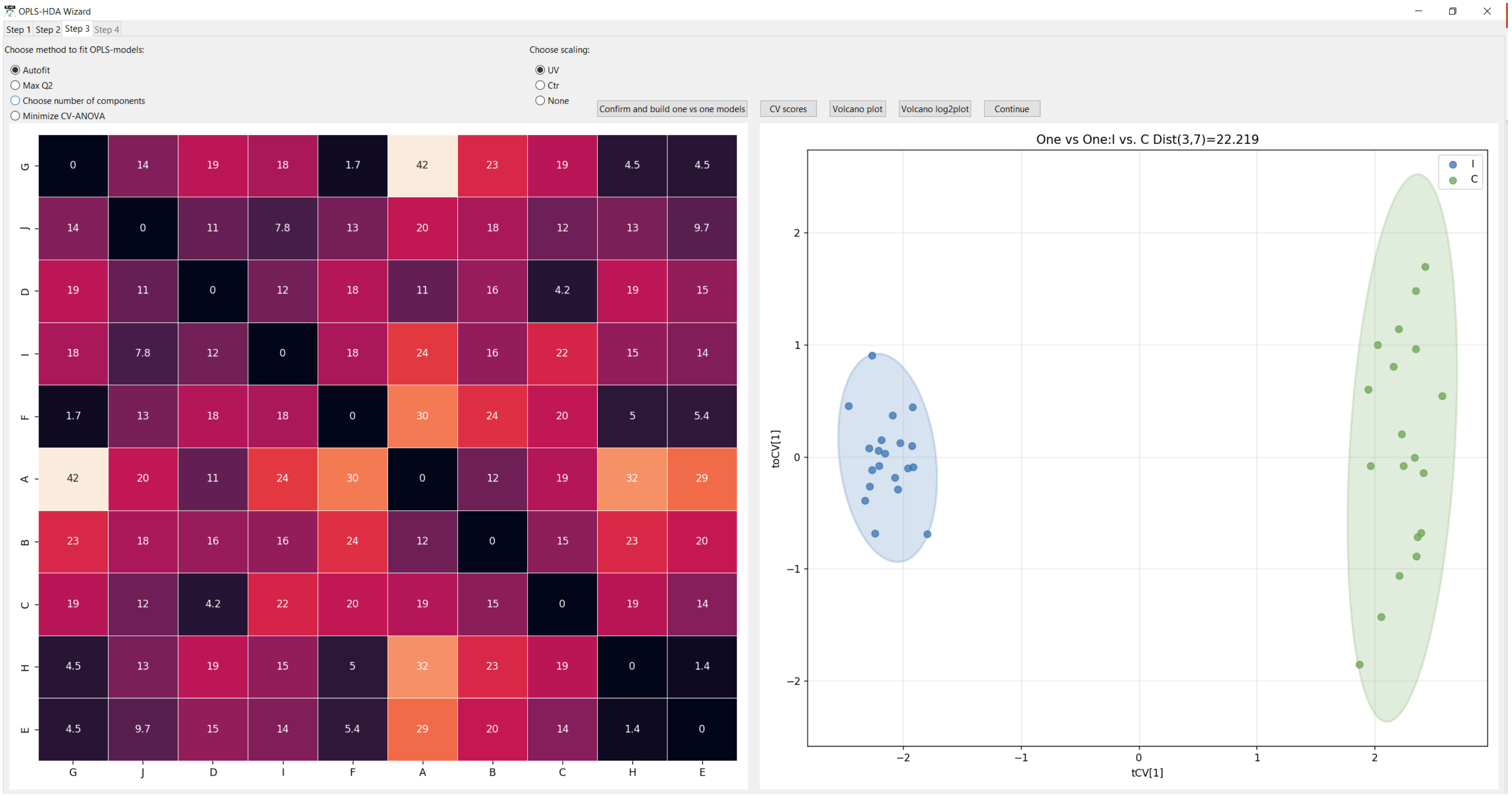
Choice 2 is slower than Choice 1 and often results in more components.
Choice 3 is fast and provides a fixed number of components.
Choice 4 is similar to Choice 2 but more restrictive regarding the number of components.

2. Choice of scaling:

- UV
- Ctr
- None

3. Confirm and build one vs. one models.

Models for all pairwise combinations of classes are computed. The distances between classes are shown in the heatmap. These distances are later used to create a dendrogram mapping the relationships between classes.

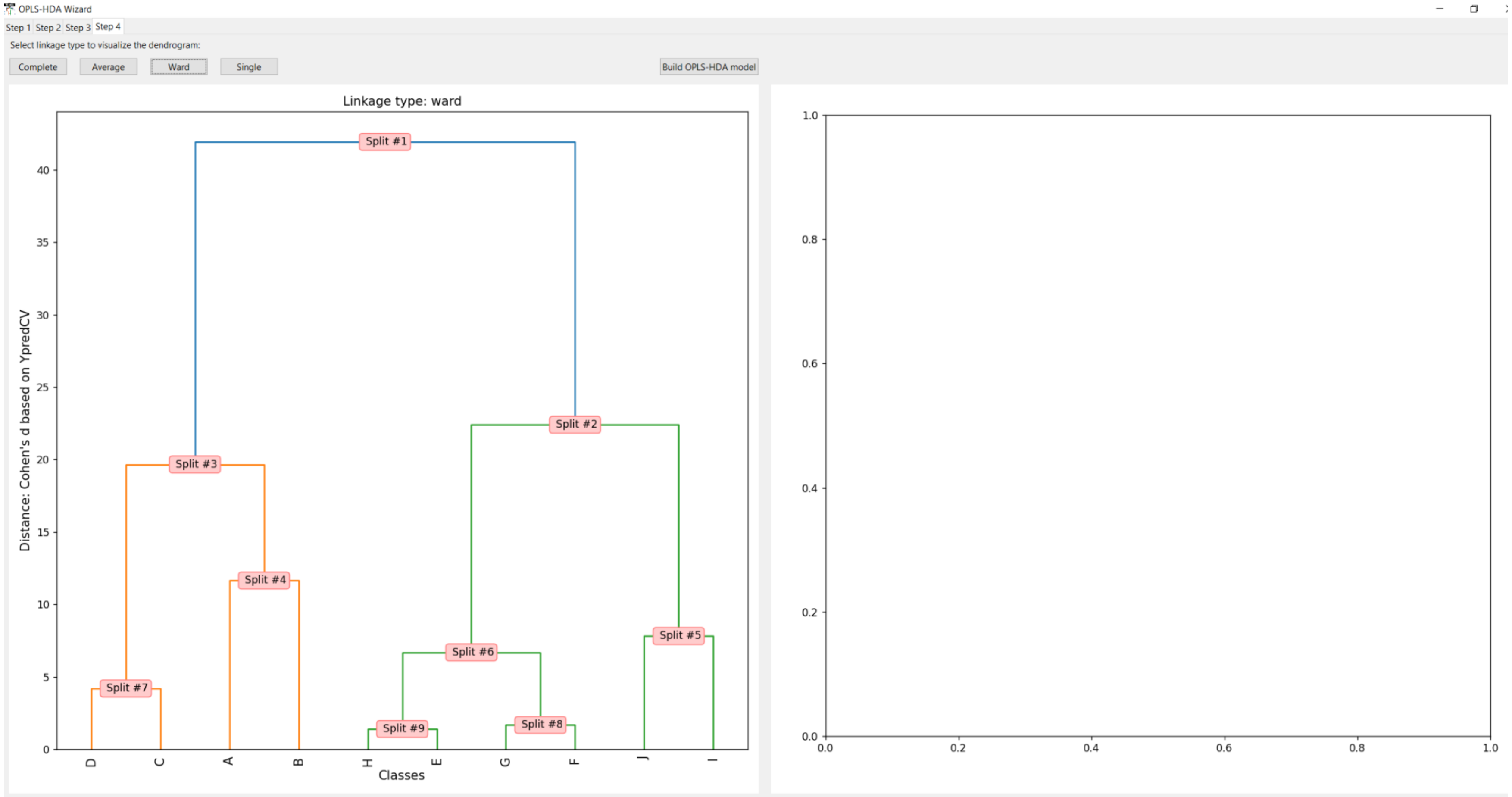


When clicking on the heatmap, the corresponding score plot will appear in the right subplot.
When done click “Continue”

Number	Model	Type	A	N × K	R2X(cum)	R2Y(cum)	Q2(cum)	Date	Title	Hierarchical
1	M1	OPLS-DA	1+0+0	43 × 9	0,804	0,982	0,98	2025-01-21	One vs One:G vs. J Dist(0,1)=13.763	
2	M2	OPLS-DA	1+0+0	42 × 9	0,89	0,991	0,99	2025-01-21	One vs One:G vs. D Dist(0,2)=19.344	
3	M3	OPLS-DA	1+1+0	40 × 9	0,9	0,991	0,988	2025-01-21	One vs One:G vs. I Dist(0,3)=17.651	
4	M4	OPLS-DA	1+1+0	41 × 9	0,555	0,612	0,413	2025-01-21	One vs One:G vs. F Dist(0,4)=1.680	
5	M5	OPLS-DA	1+0+0	40 × 9	0,873	0,998	0,998	2025-01-21	One vs One:G vs. A Dist(0,5)=41.818	
6	M6	OPLS-DA	1+0+0	42 × 9	0,921	0,994	0,993	2025-01-21	One vs One:G vs. B Dist(0,6)=22.597	
7	M7	OPLS-DA	1+0+0	39 × 9	0,726	0,991	0,989	2025-01-21	One vs One:G vs. C Dist(0,7)=18.953	
8	M8	OPLS-DA	1+0+0	43 × 9	0,429	0,855	0,836	2025-01-21	One vs One:G vs. H Dist(0,8)=4.450	
9	M9	OPLS-DA	1+0+0	40 × 9	0,387	0,867	0,84	2025-01-21	One vs One:G vs. E Dist(0,9)=4.471	
10	M10	OPLS-DA	1+1+0	45 × 9	0,893	0,971	0,969	2025-01-21	One vs One:J vs. D Dist(1,2)=10.955	
11	M11	OPLS-DA	1+1+0	43 × 9	0,908	0,947	0,941	2025-01-21	One vs One:J vs. I Dist(1,3)=7.815	
12	M12	OPLS-DA	1+0+0	44 × 9	0,795	0,978	0,977	2025-01-21	One vs One:J vs. F Dist(1,4)=12.833	
13	M13	OPLS-DA	1+0+0	43 × 9	0,989	0,991	0,991	2025-01-21	One vs One:J vs. A Dist(1,5)=20.037	
14	M14	OPLS-DA	1+1+0	45 × 9	0,956	0,99	0,988	2025-01-21	One vs One:J vs. B Dist(1,6)=17.696	
15	M15	OPLS-DA	1+0+0	42 × 9	0,605	0,977	0,973	2025-01-21	One vs One:J vs. C Dist(1,7)=11.784	
16	M16	OPLS-DA	1+0+0	46 × 9	0,767	0,981	0,979	2025-01-21	One vs One:J vs. H Dist(1,8)=13.222	
17	M17	OPLS-DA	1+0+0	43 × 9	0,728	0,964	0,961	2025-01-21	One vs One:J vs. E Dist(1,9)=9.717	
18	M18	OPLS-DA	1+1+0	42 × 9	0,969	0,984	0,975	2025-01-21	One vs One:D vs. I Dist(2,3)=12.312	
19	M19	OPLS-DA	1+0+0	43 × 9	0,869	0,989	0,989	2025-01-21	One vs One:D vs. F Dist(2,4)=18.484	
20	M20	OPLS-DA	1+1+0	42 × 9	0,885	0,984	0,966	2025-01-21	One vs One:D vs. A Dist(2,5)=10.653	
21	M21	OPLS-DA	1+1+0	44 × 9	0,818	0,988	0,985	2025-01-21	One vs One:D vs. B Dist(2,6)=15.847	
22	M22	OPLS-DA	1+1+0	41 × 9	0,842	0,847	0,821	2025-01-21	One vs One:D vs. C Dist(2,7)=4.195	
23	M23	OPLS-DA	1+0+0	45 × 9	0,805	0,99	0,99	2025-01-21	One vs One:D vs. H Dist(2,8)=19.411	
24	M24	OPLS-DA	1+0+0	42 × 9	0,766	0,985	0,984	2025-01-21	One vs One:D vs. E Dist(2,9)=15.406	
25	M25	OPLS-DA	1+1+0	41 × 9	0,856	0,99	0,988	2025-01-21	One vs One:I vs. F Dist(3,4)=17.770	
26	M26	OPLS-DA	1+0+0	40 × 9	0,993	0,994	0,994	2025-01-21	One vs One:I vs. A Dist(3,5)=24.413	
27	M27	OPLS-DA	1+1+0	42 × 9	0,976	0,987	0,984	2025-01-21	One vs One:I vs. B Dist(3,6)=15.579	
28	M28	OPLS-DA	1+3+0	39 × 9	0,99	0,995	0,992	2025-01-21	One vs One:I vs. C Dist(3,7)=22.219	
29	M29	OPLS-DA	1+1+0	43 × 9	0,937	0,985	0,984	2025-01-21	One vs One:I vs. H Dist(3,8)=15.419	
30	M30	OPLS-DA	1+1+0	40 × 9	0,914	0,983	0,98	2025-01-21	One vs One:I vs. E Dist(3,9)=13.704	
31	M31	OPLS-DA	1+0+0	41 × 9	0,848	0,996	0,996	2025-01-21	One vs One:F vs. A Dist(4,5)=30.077	
32	M32	OPLS-DA	1+0+0	43 × 9	0,912	0,994	0,993	2025-01-21	One vs One:F vs. B Dist(4,6)=23.560	
33	M33	OPLS-DA	1+0+0	40 × 9	0,704	0,992	0,99	2025-01-21	One vs One:F vs. C Dist(4,7)=19.917	
34	M34	OPLS-DA	1+0+0	44 × 9	0,416	0,881	0,867	2025-01-21	One vs One:F vs. H Dist(4,8)=5.002	
35	M35	OPLS-DA	1+2+0	41 × 9	0,78	0,911	0,885	2025-01-21	One vs One:F vs. E Dist(4,9)=5.432	
36	M36	OPLS-DA	1+0+0	42 × 9	0,957	0,975	0,973	2025-01-21	One vs One:A vs. B Dist(5,6)=11.650	
37	M37	OPLS-DA	1+1+0	39 × 9	0,961	0,99	0,989	2025-01-21	One vs One:A vs. C Dist(5,7)=18.699	
38	M38	OPLS-DA	1+1+0	43 × 9	0,991	0,997	0,996	2025-01-21	One vs One:A vs. H Dist(5,8)=32.344	
39	M39	OPLS-DA	1+1+0	40 × 9	0,989	0,996	0,995	2025-01-21	One vs One:A vs. E Dist(5,9)=28.762	
40	M40	OPLS-DA	1+1+0	41 × 9	0,8	0,985	0,982	2025-01-21	One vs One:B vs. C Dist(6,7)=14.569	
41	M41	OPLS-DA	1+0+0	45 × 9	0,823	0,993	0,993	2025-01-21	One vs One:B vs. H Dist(6,8)=22.728	
42	M42	OPLS-DA	1+0+0	42 × 9	0,809	0,991	0,991	2025-01-21	One vs One:B vs. E Dist(6,9)=20.280	
43	M43	OPLS-DA	1+1+0	42 × 9	0,826	0,991	0,99	2025-01-21	One vs One:C vs. H Dist(7,8)=19.197	
44	M44	OPLS-DA	1+0+0	39 × 9	0,647	0,983	0,981	2025-01-21	One vs One:C vs. E Dist(7,9)=14.127	
45	M45	OPLS-DA	1+1+0	43 × 9	0,377	0,466	0,33	2025-01-21	One vs One:H vs. E Dist(8,9)=1.401	

All pairwise models are created in Simca and can be seen in the project window. They are named as follows:
One vs. One X vs. Y Dist(a, b) = d

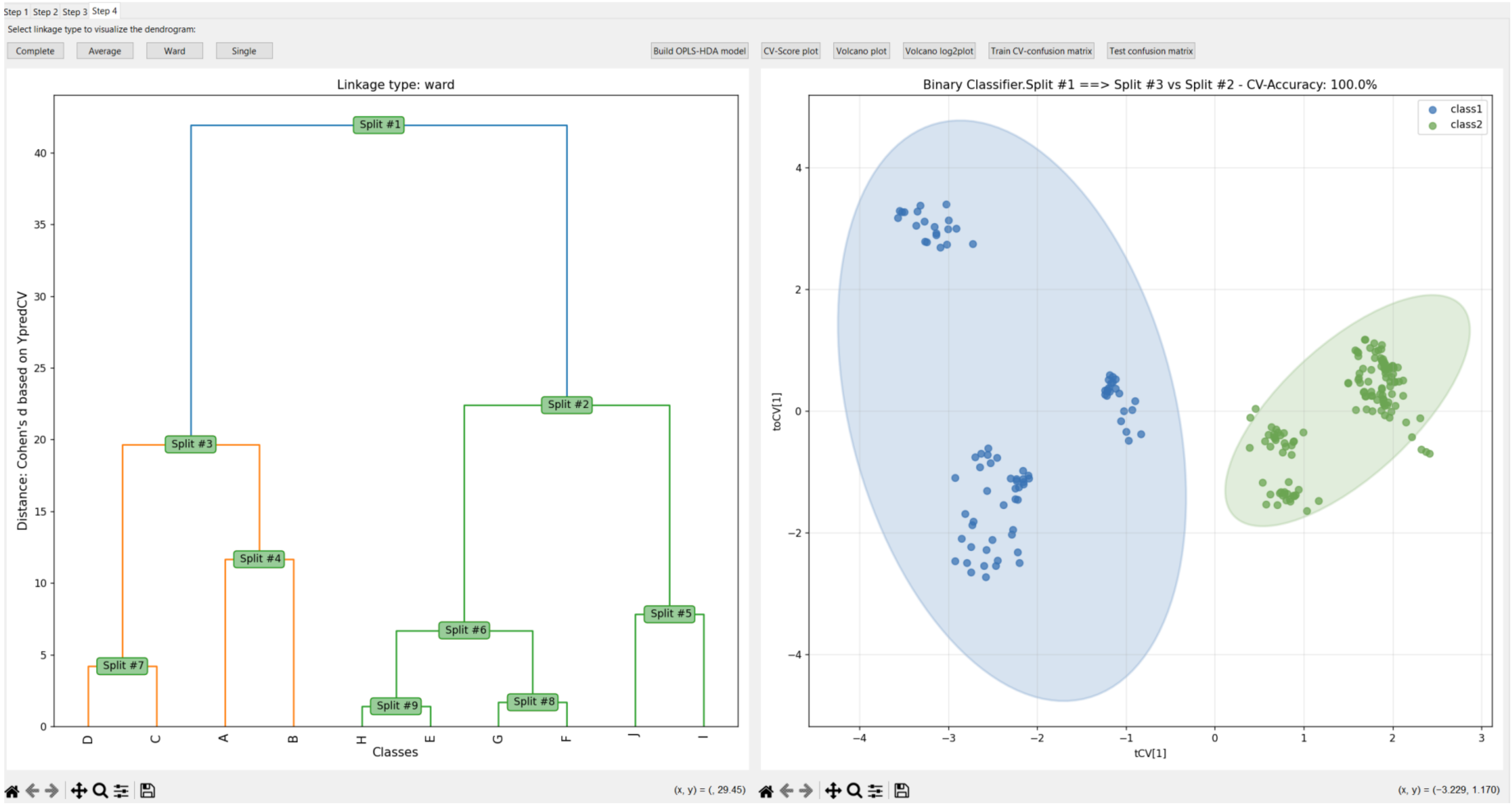
- "X" = name of class "a"
- "Y" = name of class "b"
- "a" and "b" are class indexes, and "d" is the distance between class "X" and "Y".



Select a linkage method using any of the four buttons: "Complete," "Average," "Ward," or "Single." A dendrogram will appear. Once the selection of the dendrogram is done, press "Build OPLS-HDA model." The red boxes with "split number" will shift to green after the OPLS-DA model in the split is calculated. The models are stored in the simca project, the models are named: Binary Classifier. Split #number ==> "left class" vs "right class" - CV-Accuracy: XX.X%

By clicking a green box, a score plot of the corresponding OPLS-DA model will be shown in the right subplot.

By clicking any of the buttons "Train CV confusion matrix" or "Test confusion matrix" (only appear if you have a test set) the confusion matrix will be shown.

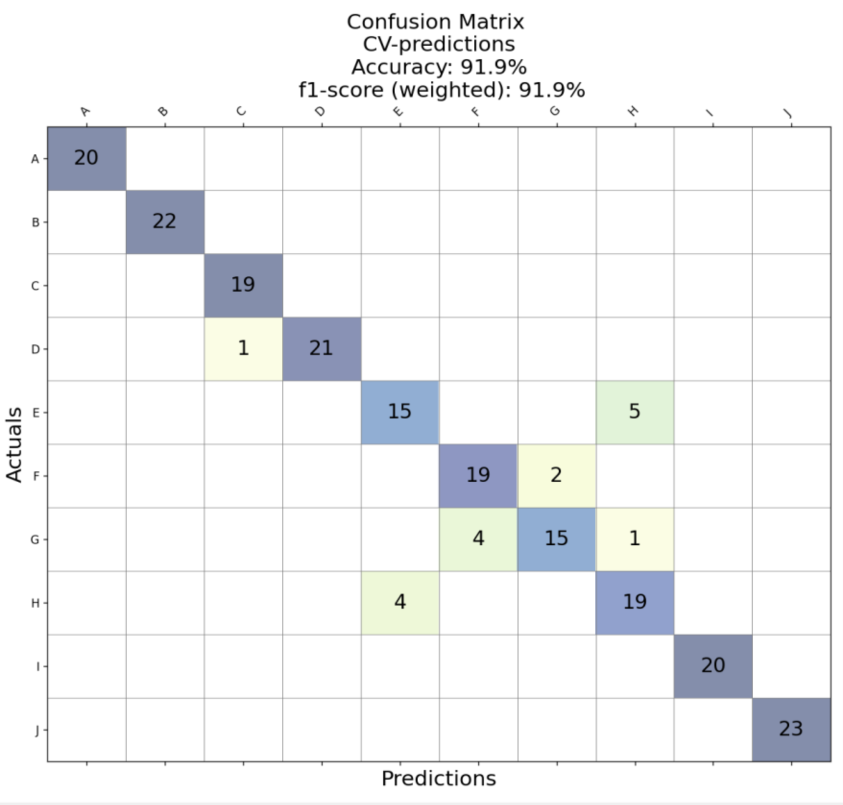


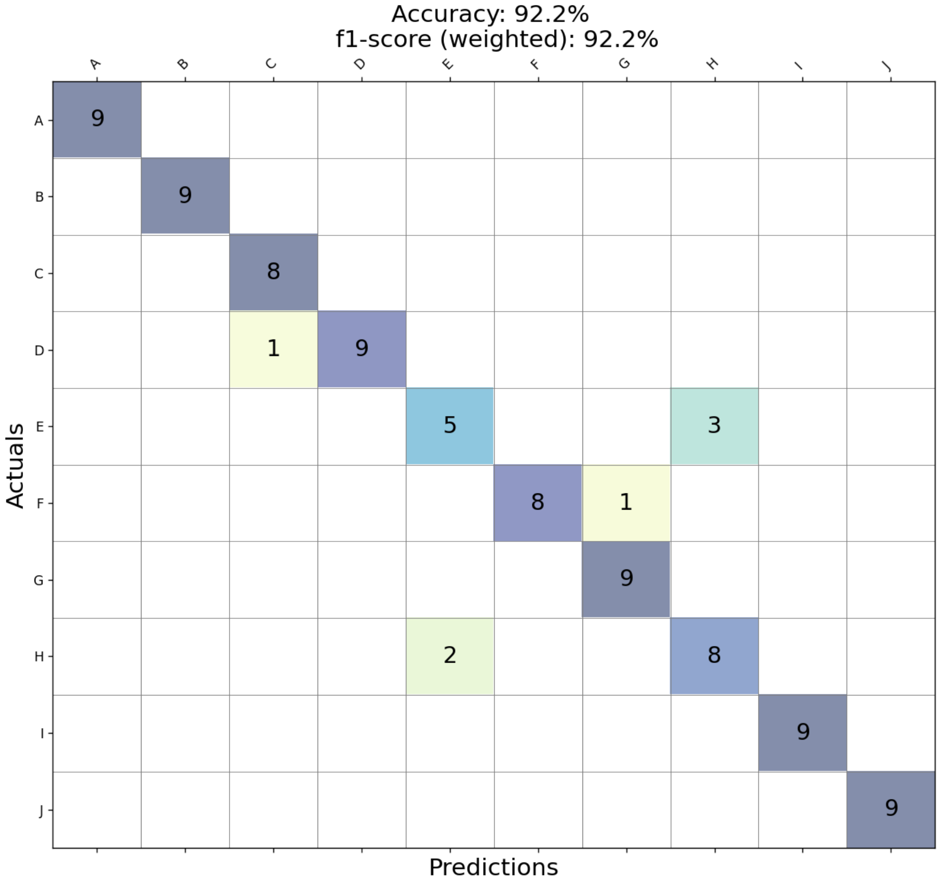
Pizza [M23]

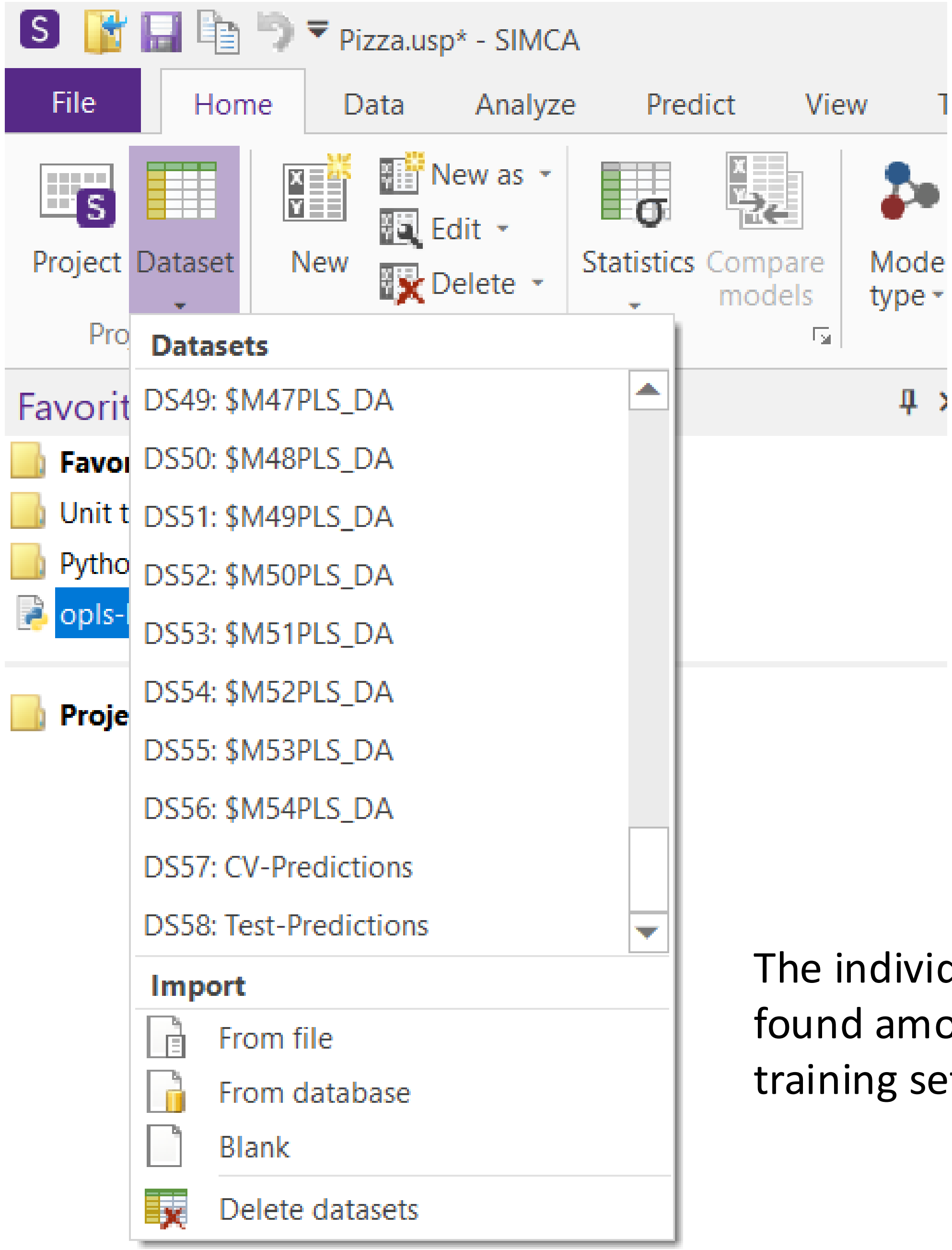
Number	Model	Type	A	N × K	R2X(cum)	R2Y(cum)	Q2(cum)	Date	Title	Hierarchical
46	M46	OPLS-DA	1+2+0	210 × 9	0,982	0,886	0,885	2025-01-21	Binary Classifier.Split #1...	
47	M47	OPLS-DA	1+1+0	127 × 9	0,883	0,967	0,965	2025-01-21	Binary Classifier.Split #2...	
48	M48	OPLS-DA	1+2+0	83 × 9	0,96	0,946	0,94	2025-01-21	Binary Classifier.Split #3...	
49	M49	OPLS-DA	1+0+0	42 × 9	0,957	0,975	0,973	2025-01-21	Binary Classifier.Split #4...	
50	M50	OPLS-DA	1+1+0	43 × 9	0,908	0,947	0,941	2025-01-21	Binary Classifier.Split #5...	
51	M51	OPLS-DA	1+0+0	84 × 9	0,339	0,862	0,854	2025-01-21	Binary Classifier.Split #6...	
52	M52	OPLS-DA	1+1+0	41 × 9	0,842	0,847	0,821	2025-01-21	Binary Classifier.Split #7...	
53	M53	OPLS-DA	1+1+0	41 × 9	0,555	0,612	0,413	2025-01-21	Binary Classifier.Split #8...	
54	M54	OPLS-DA	1+1+0	43 × 9	0,377	0,466	0,33	2025-01-21	Binary Classifier.Split #9...	

Train CV-confusion matrix

Test confusion matrix





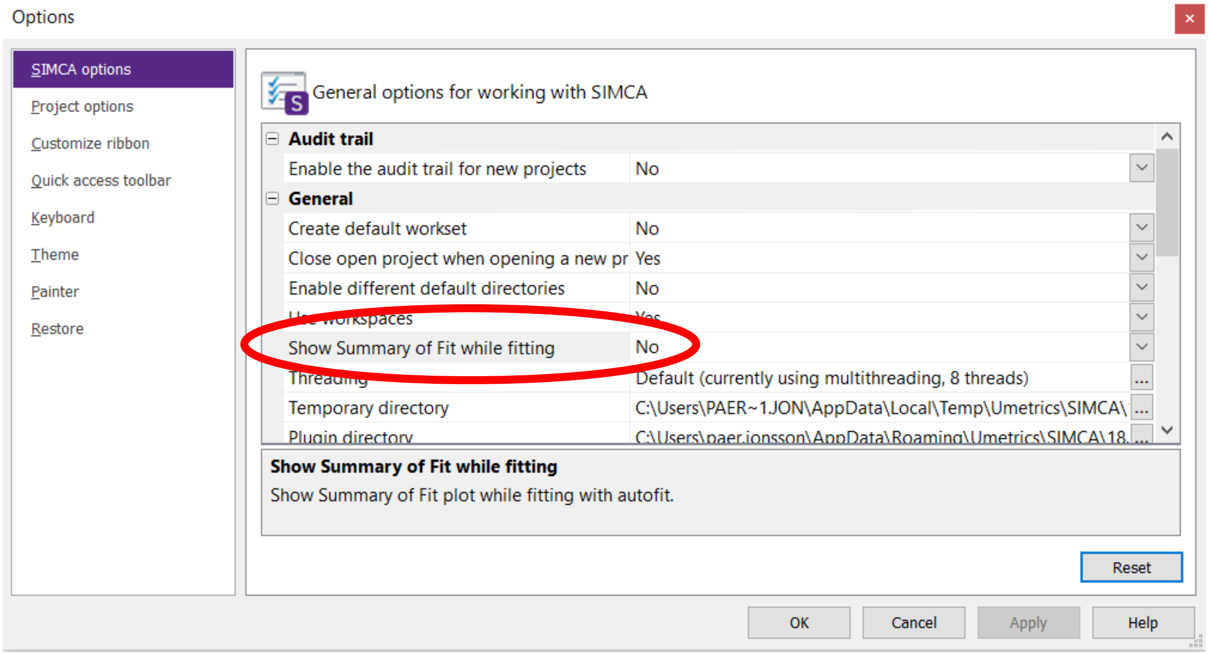


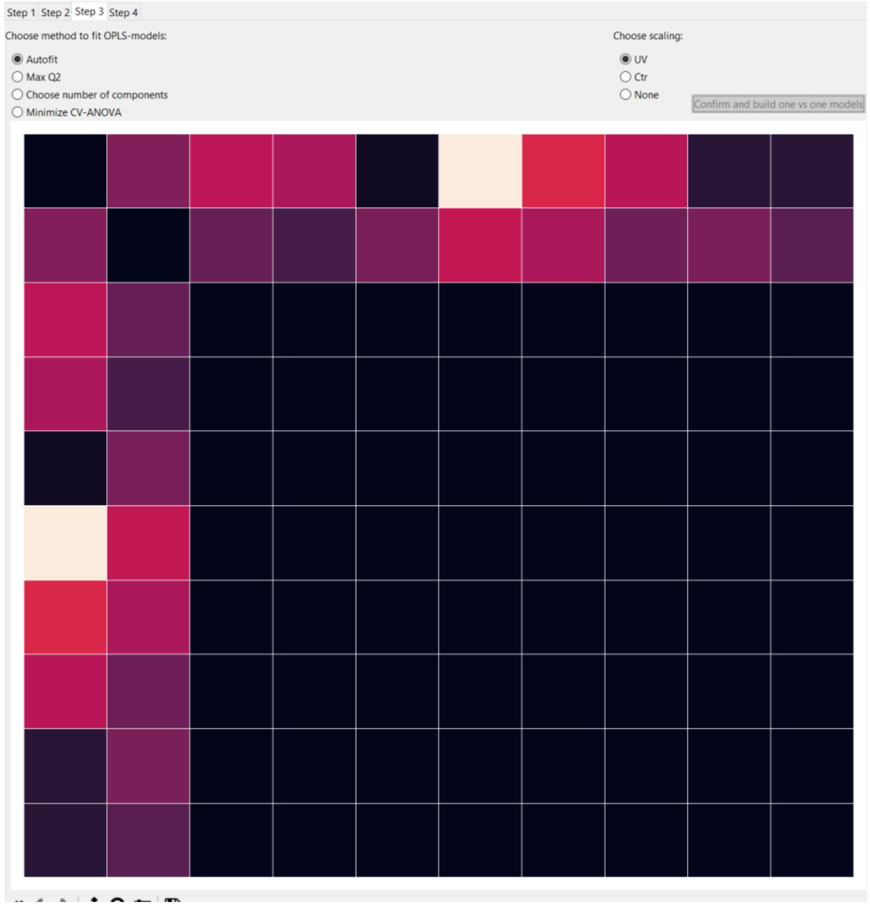
Test-Predictions

	1	2	3	4	5	6
1	Primary ID	Class PS	Predicted PS	Correct prediction PS	Max DModPS PS	Max DModXPS - Split # PS
2	7	A	A	yes	1,00052	4
3	8	A	A	yes	2,18594	4
4	11	A	A	yes	0,828876	3
5	15	A	A	yes	0,648385	4
6	19	A	A	yes	1,52732	4
7	20	A	A	yes	1,39871	4
8	21	A	A	yes	1,17335	4
9	22	A	A	yes	0,917547	3
10	27	A	A	yes	0,881637	4
11	39	B	B	yes	1,83986	4
12	41	B	B	yes	1,29274	1
13	44	B	B	yes	1,22528	4
14	45	B	B	yes	1,73083	4
15	48	B	B	yes	0,755464	3
16	52	B	B	yes	1,04668	4
17	56	B	B	yes	0,793195	3
18	57	B	B	yes	1,15264	4
19	59	B	B	yes	2,39825	1
20	63	C	C	yes	1,09339	1
21	64	C	C	yes	0,55305	1

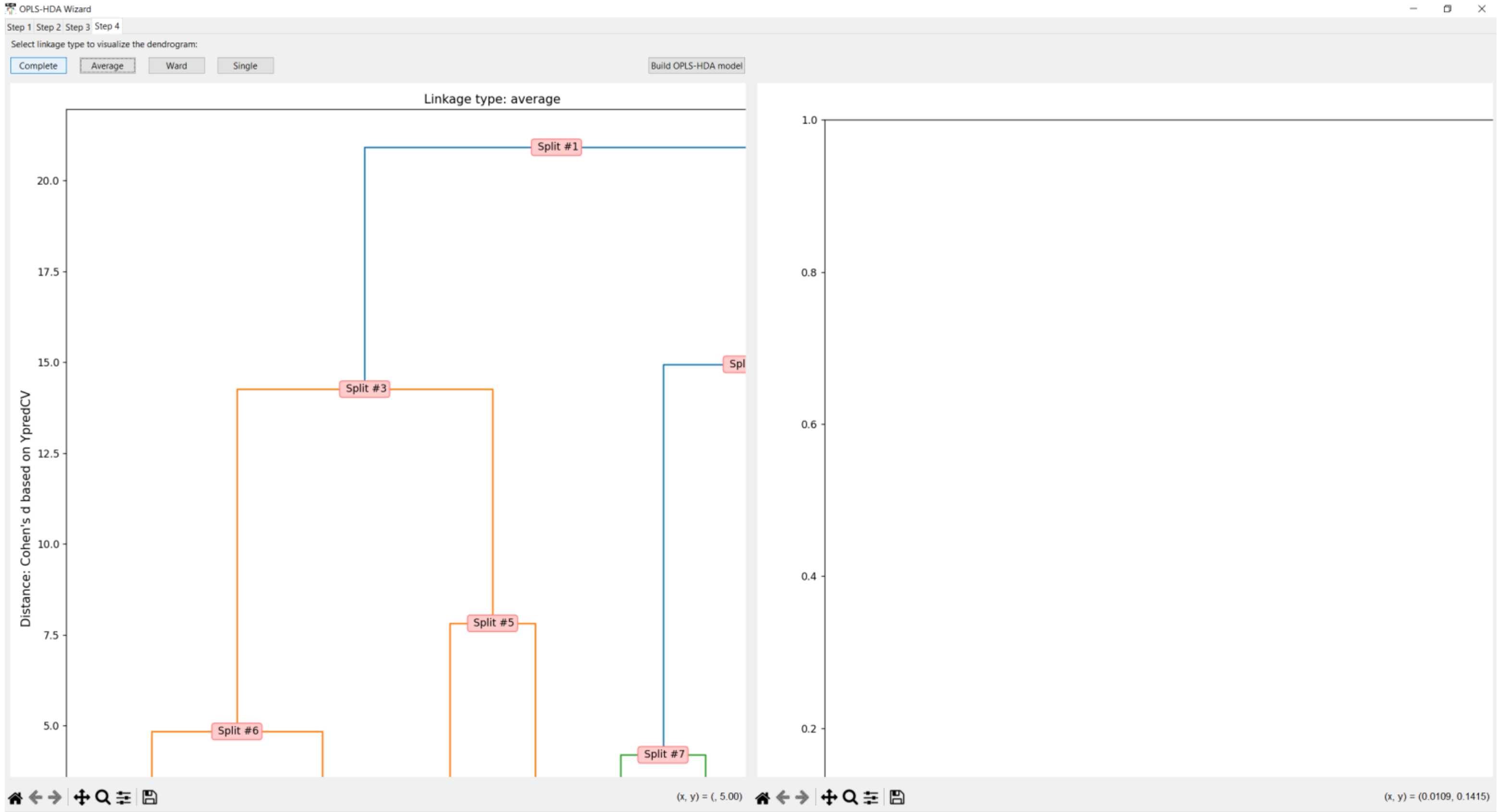
The individual predictions of observations in "training set" and "test set" can be found among "Dataset" in Simca. Cross-validated predictions are shown for the training set.

When using datasets with a large number of classes, it is advisable to set “**Show Summary of Fit While Fitting**” to “**No**”.





During calculations of "one vs. one" models, the update of the heatmap may stop if the GUI window is moved, resized, clicked on, or due to heavy calculations with large data sets.
Solution: Wait.



If figures do not fit inside the subplot, press "R" to resize the GUI window. Do it twice to maintain the size of the GUI window.